

Mathematical Foundations of Machine Learning

Assignment 3 Solutions

Question 1 (20 points)

Let $V = \mathbb{R}^3$ with ordered bases

$$B = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3), \quad \mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix},$$

and

$$\tilde{B} = (\tilde{\mathbf{b}}_1, \tilde{\mathbf{b}}_2, \tilde{\mathbf{b}}_3), \quad \tilde{\mathbf{b}}_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \quad \tilde{\mathbf{b}}_2 = \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}, \quad \tilde{\mathbf{b}}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

Let $W = \mathbb{R}^2$ with ordered bases

$$C = (\mathbf{c}_1, \mathbf{c}_2), \quad \mathbf{c}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \tilde{C} = (\tilde{\mathbf{c}}_1, \tilde{\mathbf{c}}_2), \quad \tilde{\mathbf{c}}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \tilde{\mathbf{c}}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A linear map $\Phi : V \rightarrow W$ has matrix (with respect to B and C)

$$[\Phi]_B^C = \mathbf{A} = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 2 \end{bmatrix}.$$

1. **(6 points)** Let $\mathbf{S} \in \mathbb{R}^{3 \times 3}$ be the coordinate-change matrix that converts $\tilde{B}$ -coordinates to B -coordinates, i.e., for all $\mathbf{v} \in V$,

$$[\mathbf{v}]_B = \mathbf{S} [\mathbf{v}]_{\tilde{B}}.$$

Compute $\mathbf{S}$. Similarly, let $\mathbf{T} \in \mathbb{R}^{2 \times 2}$ convert $\tilde{C}$ -coordinates to C -coordinates, i.e., for all $\mathbf{w} \in W$,

$$[\mathbf{w}]_C = \mathbf{T} [\mathbf{w}]_{\tilde{C}}.$$

Compute $\mathbf{T}$.

2. **(6 points)** Let $\tilde{\mathbf{A}}$ be the matrix of Φ with respect to $\tilde{B}$ and $\tilde{C}$, i.e.,

$$[\Phi(\mathbf{v})]_{\tilde{C}} = \tilde{\mathbf{A}} [\mathbf{v}]_{\tilde{B}} \quad \text{for all } \mathbf{v} \in V.$$

Compute $\tilde{\mathbf{A}}$.

3. (8 points) Let $[\mathbf{x}]_{\tilde{B}} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$. Compute $\Phi(\mathbf{x})$ in *standard coordinates* of $\mathbb{R}^2$.

Solution.

1. By definition, the columns of $\mathbf{S}$ are the B -coordinate vectors of $\tilde{\mathbf{b}}_1, \tilde{\mathbf{b}}_2, \tilde{\mathbf{b}}_3$.
Observe:

$$\tilde{\mathbf{b}}_1 = \mathbf{b}_1 + \mathbf{b}_2, \quad \tilde{\mathbf{b}}_2 = \mathbf{b}_2 + \mathbf{b}_3, \quad \tilde{\mathbf{b}}_3 = \mathbf{b}_1 + \mathbf{b}_3.$$

Hence

$$\mathbf{S} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Similarly, the columns of $\mathbf{T}$ are the C -coordinates of $\tilde{\mathbf{c}}_1, \tilde{\mathbf{c}}_2$. Since

$$2\mathbf{c}_1 - \mathbf{c}_2 = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \tilde{\mathbf{c}}_1, \quad \tilde{\mathbf{c}}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \mathbf{c}_1,$$

we get

$$\mathbf{T} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}.$$

2. The change-of-basis relation gives

$$[\Phi(\mathbf{v})]_C = \mathbf{A}[\mathbf{v}]_B, \quad [\mathbf{v}]_B = \mathbf{S}[\mathbf{v}]_{\tilde{B}}, \quad [\Phi(\mathbf{v})]_C = \mathbf{T}[\Phi(\mathbf{v})]_{\tilde{C}}.$$

Therefore,

$$\mathbf{T}[\Phi(\mathbf{v})]_{\tilde{C}} = \mathbf{A}\mathbf{S}[\mathbf{v}]_{\tilde{B}} \implies [\Phi(\mathbf{v})]_{\tilde{C}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{S}[\mathbf{v}]_{\tilde{B}},$$

so

$$\tilde{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{S}.$$

First compute

$$\mathbf{T}^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}.$$

Then

$$\tilde{\mathbf{A}} = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -4 & -3 & -5 \\ 7 & 4 & 11 \end{bmatrix}.$$

3. Convert $\tilde{B}$ -coordinates to B -coordinates:

$$[\mathbf{x}]_B = \mathbf{S}[\mathbf{x}]_{\tilde{B}} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}.$$

Compute C -coordinates:

$$[\Phi(\mathbf{x})]_C = \mathbf{A}[\mathbf{x}]_B = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 20 \end{bmatrix}.$$

Convert to standard coordinates using $\mathbf{c}_1, \mathbf{c}_2$:

$$\Phi(\mathbf{x}) = 3\mathbf{c}_1 + 20\mathbf{c}_2 = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 20 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 43 \\ 23 \end{bmatrix}.$$

Question 2 (22 points)

Let $\mathbf{A} \in \mathbb{R}^{3 \times 4}$ and $L: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be given by $L(\mathbf{x}) = \mathbf{A}\mathbf{x}$:

$$\mathbf{A} = \begin{bmatrix} -1 & 2 & -2 & -1 \\ 2 & -2 & 2 & -2 \\ -2 & 2 & -2 & 2 \end{bmatrix}.$$

- (10 points) Determine $\text{rank}(\mathbf{A})$ and a basis for $\text{Im}(L)$. Then find a nonzero $\mathbf{y} \in \mathbb{R}^3$ such that $\mathbf{y}^\top \mathbf{w} = 0$ for all $\mathbf{w} \in \text{Im}(L)$.
- (12 points) Determine whether the system $L(\mathbf{x}) = \mathbf{0}$ has a unique solution. If it does, find it. If it does not, find a basis for $\text{Ker}(L)$ and describe the full solution set.

Solution.

- Row-reduce:

$$\begin{aligned} & \begin{bmatrix} -1 & 2 & -2 & -1 \\ 2 & -2 & 2 & -2 \\ -2 & 2 & -2 & 2 \end{bmatrix} \xrightarrow{R_1 \leftarrow -R_1} \begin{bmatrix} 1 & -2 & 2 & 1 \\ 2 & -2 & 2 & -2 \\ -2 & 2 & -2 & 2 \end{bmatrix} \\ & \xrightarrow{R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 + 2R_1} \begin{bmatrix} 1 & -2 & 2 & 1 \\ 0 & 2 & -2 & -4 \\ 0 & -2 & 2 & 4 \end{bmatrix} \\ & \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{bmatrix} 1 & -2 & 2 & 1 \\ 0 & 2 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Continue to RREF:

$$\begin{bmatrix} 1 & -2 & 2 & 1 \\ 0 & 2 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} \begin{bmatrix} 1 & -2 & 2 & 1 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are 2 pivots, so $\text{rank}(\mathbf{A}) = 2$, with pivot columns 1, 2. Hence a basis for $\text{Im}(L)$ is given by columns 1, 2 of the *original* matrix:

$$\text{Im}(L) = \text{span} \left\{ \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} \right\}.$$

To find $\mathbf{y} \neq \mathbf{0}$ such that $\mathbf{y}^\top \mathbf{w} = 0$ for all $\mathbf{w} \in \text{Im}(L)$, it suffices to choose $\mathbf{y}$ with $\mathbf{A}^\top \mathbf{y} = \mathbf{0}$. One such solution is

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

since

$$\mathbf{A}^\top \mathbf{y} = \begin{bmatrix} -1 & 2 & -2 \\ 2 & -2 & 2 \\ -2 & 2 & -2 \\ -1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

2. The system $L(\mathbf{x}) = \mathbf{0}$ is $\mathbf{A}\mathbf{x} = \mathbf{0}$, i.e. $\mathbf{x} \in \text{Ker}(L)$. It has a unique solution iff $\text{Ker}(L) = \{\mathbf{0}\}$.

From the RREF equations:

$$x_1 - 3x_4 = 0, \quad x_2 - x_3 - 2x_4 = 0,$$

take free variables $x_3 = s$ and $x_4 = t$. Then

$$x_1 = 3t, \quad x_2 = s + 2t,$$

so

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$$

Since there are nonzero solutions (e.g. take $s = 1, t = 0$), the solution is *not* unique. A basis for $\text{Ker}(L)$ is

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\},$$

and the full solution set is

$$\text{Ker}(L) = \left\{ s \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix} : s, t \in \mathbb{R} \right\}.$$

Question 3 (28 points)

Let

$$\mathbf{M} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \in \mathbb{R}^{3 \times 3}, \quad \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \mathbf{x}^\top \mathbf{M} \mathbf{y}.$$

1. **(8 points)** Prove that $\mathbf{M}$ is positive definite by showing $\mathbf{x}^\top \mathbf{M} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
2. **(8 points)** Prove that $\langle \cdot, \cdot \rangle_{\mathbf{M}}$ is an inner product on $\mathbb{R}^3$.
3. **(6 points)** Let $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Compute $\cos \theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$ under $\langle \cdot, \cdot \rangle_{\mathbf{M}}$.
4. **(6 points)** Let $\mathbf{w}(t) = \begin{bmatrix} 1 \\ t \\ 0 \end{bmatrix}$. Find all real t such that the angle between $\mathbf{u}$ and $\mathbf{w}(t)$ under $\langle \cdot, \cdot \rangle_{\mathbf{M}}$ equals 60° .

Solution.

1. Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Then

$$\mathbf{x}^\top \mathbf{M} \mathbf{x} = 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_2x_3.$$

Rewrite as a sum of squares:

$$\begin{aligned} (x_1 + x_2)^2 + (x_2 + x_3)^2 + x_1^2 + x_3^2 &= (x_1^2 + 2x_1x_2 + x_2^2) + (x_2^2 + 2x_2x_3 + x_3^2) + x_1^2 + x_3^2 \\ &= 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_2x_3 = \mathbf{x}^\top \mathbf{M} \mathbf{x}. \end{aligned}$$

If $\mathbf{x} \neq \mathbf{0}$, then at least one of x_1, x_2, x_3 is nonzero, hence the sum of squares

$$(x_1 + x_2)^2 + (x_2 + x_3)^2 + x_1^2 + x_3^2$$

is strictly positive. Therefore $\mathbf{x}^\top \mathbf{M} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$, so $\mathbf{M}$ is positive definite.

2. We check the inner product axioms. For all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$:

$$\langle \alpha \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle_{\mathbf{M}} = (\alpha \mathbf{x} + \mathbf{y})^\top \mathbf{M} \mathbf{z} = \alpha \mathbf{x}^\top \mathbf{M} \mathbf{z} + \mathbf{y}^\top \mathbf{M} \mathbf{z} = \alpha \langle \mathbf{x}, \mathbf{z} \rangle_{\mathbf{M}} + \langle \mathbf{y}, \mathbf{z} \rangle_{\mathbf{M}},$$

so it is linear in the first argument. Since $\mathbf{M}$ is symmetric,

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \mathbf{x}^\top \mathbf{M} \mathbf{y} = (\mathbf{y}^\top \mathbf{M} \mathbf{x})^\top = \mathbf{y}^\top \mathbf{M}^\top \mathbf{x} = \mathbf{y}^\top \mathbf{M} \mathbf{x} = \langle \mathbf{y}, \mathbf{x} \rangle_{\mathbf{M}},$$

so it is symmetric. Finally,

$$\langle \mathbf{x}, \mathbf{x} \rangle_{\mathbf{M}} = \mathbf{x}^\top \mathbf{M} \mathbf{x} > 0 \quad \text{for all } \mathbf{x} \neq \mathbf{0},$$

by part (1), and equals 0 when $\mathbf{x} = \mathbf{0}$. Hence $\langle \cdot, \cdot \rangle_{\mathbf{M}}$ is an inner product.

3. Compute:

$$\langle \mathbf{u}, \mathbf{u} \rangle_M = \mathbf{u}^\top \mathbf{M} \mathbf{u} = 4, \quad \langle \mathbf{v}, \mathbf{v} \rangle_M = \mathbf{v}^\top \mathbf{M} \mathbf{v} = 6, \quad \langle \mathbf{u}, \mathbf{v} \rangle_M = \mathbf{u}^\top \mathbf{M} \mathbf{v} = 4.$$

Therefore

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle_M}{\|\mathbf{u}\|_M \|\mathbf{v}\|_M} = \frac{4}{\sqrt{4} \sqrt{6}} = \frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3}.$$

4. We need

$$\frac{\langle \mathbf{u}, \mathbf{w}(t) \rangle_M}{\|\mathbf{u}\|_M \|\mathbf{w}(t)\|_M} = \cos 60^\circ = \frac{1}{2}.$$

Compute

$$\langle \mathbf{u}, \mathbf{w}(t) \rangle_M = \mathbf{u}^\top \mathbf{M} \mathbf{w}(t) = 2t+2, \quad \langle \mathbf{w}(t), \mathbf{w}(t) \rangle_M = \mathbf{w}(t)^\top \mathbf{M} \mathbf{w}(t) = 2(t^2+t+1),$$

and $\|\mathbf{u}\|_M = \sqrt{4} = 2$. So the equation becomes

$$\frac{2t+2}{2\sqrt{2(t^2+t+1)}} = \frac{1}{2} \implies \frac{t+1}{\sqrt{2(t^2+t+1)}} = \frac{1}{2}.$$

Since $\cos 60^\circ > 0$, we must have $t+1 > 0$. Squaring gives

$$4(t+1)^2 = 2(t^2+t+1) \implies 2(t+1)^2 = t^2+t+1 \implies t^2+3t+1=0.$$

Thus

$$t = \frac{-3 \pm \sqrt{5}}{2}.$$

Enforcing $t+1 > 0$ leaves

$$t = \frac{-3 + \sqrt{5}}{2}.$$

Question 4 (30 points)

Let

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}.$$

- (10 points)** Compute the angle between $\mathbf{a}$ and $\mathbf{b}$ (in exact form, using $\arccos(\cdot)$).
- (10 points)** Find all real λ such that the angle between $\mathbf{a} + \lambda \mathbf{b}$ and $\mathbf{b}$ equals 45° .
- (10 points)** Prove the following: for nonzero vectors $\mathbf{x}, \mathbf{y}$ and scalars $\alpha, \beta \neq 0$, the angle between $\alpha \mathbf{x}$ and $\beta \mathbf{y}$ depends only on the sign of $\alpha\beta$, and equals the angle between $\mathbf{x}$ and $\mathbf{y}$ when $\alpha\beta > 0$.

Solution.

1. Compute dot product and norms:

$$\mathbf{a} \cdot \mathbf{b} = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = 2 - 2 - 2 = -2,$$

$$\|\mathbf{a}\| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}, \quad \|\mathbf{b}\| = \sqrt{2^2 + (-1)^2 + 2^2} = 3.$$

Hence

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{-2}{3\sqrt{6}}, \quad \theta = \arccos\left(\frac{-2}{3\sqrt{6}}\right).$$

2. Let $\mathbf{x}(\lambda) = \mathbf{a} + \lambda\mathbf{b}$. We need

$$\frac{\mathbf{x}(\lambda) \cdot \mathbf{b}}{\|\mathbf{x}(\lambda)\| \|\mathbf{b}\|} = \cos 45^\circ = \frac{\sqrt{2}}{2}.$$

Compute

$$\mathbf{x}(\lambda) \cdot \mathbf{b} = (\mathbf{a} + \lambda\mathbf{b}) \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \lambda(\mathbf{b} \cdot \mathbf{b}) = -2 + 9\lambda,$$

and

$$\|\mathbf{x}(\lambda)\|^2 = \|\mathbf{a} + \lambda\mathbf{b}\|^2 = \|\mathbf{a}\|^2 + 2\lambda(\mathbf{a} \cdot \mathbf{b}) + \lambda^2\|\mathbf{b}\|^2 = 6 + 2\lambda(-2) + 9\lambda^2 = 9\lambda^2 - 4\lambda + 6.$$

Since $\|\mathbf{b}\| = 3$, the equation is

$$\frac{9\lambda - 2}{3\sqrt{9\lambda^2 - 4\lambda + 6}} = \frac{\sqrt{2}}{2}.$$

Squaring:

$$\frac{(9\lambda - 2)^2}{9(9\lambda^2 - 4\lambda + 6)} = \frac{1}{2} \implies 2(9\lambda - 2)^2 = 9(9\lambda^2 - 4\lambda + 6).$$

Expand:

$$\begin{aligned} 2(81\lambda^2 - 36\lambda + 4) &= 81\lambda^2 - 36\lambda + 54 \\ 162\lambda^2 - 72\lambda + 8 &= 81\lambda^2 - 36\lambda + 54 \implies 81\lambda^2 - 36\lambda - 46 = 0. \end{aligned}$$

Solve:

$$\lambda = \frac{36 \pm \sqrt{36^2 + 4 \cdot 81 \cdot 46}}{2 \cdot 81} = \frac{36 \pm \sqrt{1296 + 14904}}{162} = \frac{36 \pm 90\sqrt{2}}{162} = \frac{2 \pm 5\sqrt{2}}{9}.$$

The angle 45° has positive cosine, so we must have $9\lambda - 2 > 0$, which selects

$$\lambda = \frac{2 + 5\sqrt{2}}{9}.$$

3. Let $\theta = \angle(\mathbf{x}, \mathbf{y})$, so

$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

For $\alpha, \beta \neq 0$,

$$\cos \angle(\alpha \mathbf{x}, \beta \mathbf{y}) = \frac{(\alpha \mathbf{x}) \cdot (\beta \mathbf{y})}{\|\alpha \mathbf{x}\| \|\beta \mathbf{y}\|} = \frac{\alpha \beta (\mathbf{x} \cdot \mathbf{y})}{(|\alpha| \|\mathbf{x}\|)(|\beta| \|\mathbf{y}\|)} = \frac{\alpha \beta}{|\alpha \beta|} \cdot \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \text{sign}(\alpha \beta) \cos \theta.$$

If $\alpha \beta > 0$, then $\text{sign}(\alpha \beta) = 1$ and

$$\cos \angle(\alpha \mathbf{x}, \beta \mathbf{y}) = \cos \theta,$$

so the angle is the same as $\angle(\mathbf{x}, \mathbf{y})$ (both lie in $[0, \pi]$). If $\alpha \beta < 0$, then $\cos$ changes sign, so the angle becomes $\pi - \theta$.

Bonus Question (10 points)

Let

$$\mathbf{M} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \mathbf{x}^{\top} \mathbf{M} \mathbf{y}.$$

Let $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

1. **(4 points)** Find the scalar α such that $\mathbf{u}$ and $\mathbf{v} + \alpha \mathbf{u}$ have angle 90° under $\langle \cdot, \cdot \rangle_{\mathbf{M}}$.
2. **(3 points)** Compute $\|\mathbf{u}\|_{\mathbf{M}}$ and $\|\mathbf{v} + \alpha \mathbf{u}\|_{\mathbf{M}}$ (with your α from part (1)).
3. **(3 points)** Let $\mathbf{p} = \mathbf{v} + \alpha \mathbf{u}$. Compute $\cos \angle(\mathbf{u}, \mathbf{u} + \mathbf{p})$ under $\langle \cdot, \cdot \rangle_{\mathbf{M}}$.

Solution.

1. Orthogonality under $\langle \cdot, \cdot \rangle_{\mathbf{M}}$ means

$$\langle \mathbf{u}, \mathbf{v} + \alpha \mathbf{u} \rangle_{\mathbf{M}} = 0.$$

Compute

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{M}} = \mathbf{u}^{\top} \mathbf{M} \mathbf{v} = 4, \quad \langle \mathbf{u}, \mathbf{u} \rangle_{\mathbf{M}} = \mathbf{u}^{\top} \mathbf{M} \mathbf{u} = 4.$$

Hence

$$0 = \langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{M}} + \alpha \langle \mathbf{u}, \mathbf{u} \rangle_{\mathbf{M}} = 4 + 4\alpha \implies \alpha = -1.$$

2. With $\alpha = -1$,

$$\|\mathbf{u}\|_M = \sqrt{4} = 2.$$

Let $\mathbf{p} = \mathbf{v} - \mathbf{u} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. Then

$$\|\mathbf{p}\|_M^2 = \mathbf{p}^\top \mathbf{M} \mathbf{p}.$$

Compute $\mathbf{M} \mathbf{p} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$, so

$$\|\mathbf{p}\|_M^2 = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = 2 \implies \|\mathbf{p}\|_M = \sqrt{2}.$$

3. Since $\mathbf{p} = \mathbf{v} - \mathbf{u}$, we have $\mathbf{u} + \mathbf{p} = \mathbf{v}$. Therefore

$$\cos \angle(\mathbf{u}, \mathbf{u} + \mathbf{p}) = \cos \angle(\mathbf{u}, \mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle_M}{\|\mathbf{u}\|_M \|\mathbf{v}\|_M}.$$

We already have $\langle \mathbf{u}, \mathbf{v} \rangle_M = 4$ and $\|\mathbf{u}\|_M = 2$. Also

$$\|\mathbf{v}\|_M^2 = \mathbf{v}^\top \mathbf{M} \mathbf{v} = 6 \implies \|\mathbf{v}\|_M = \sqrt{6}.$$

Hence

$$\cos \angle(\mathbf{u}, \mathbf{u} + \mathbf{p}) = \frac{4}{2\sqrt{6}} = \frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3}.$$